

Configuration List

12.5MW/50.15MWh Liquid-Cooling Energy Storage System Configuration List					South Africa
No.	Item	Specifications	Unit	Qty	Remark
1	1500V Liquid Cooling Battery Container	20 HC 5.015MWh Lithium iron phosphate battery system	set	10	
1.1	Battery	5.015MWh (314Ah, 1P104S4S*12, 0.25C, including BMU, BCU, BAU three-level battery management, high-voltage box, battery rack and system accessories, etc.)	set	1	
1.2	Control Busbar Cabinet	12 in 1, including BAU, UPS, switches, etc.	set	1	
1.3	Temperature Control System	Liquid cooling unit	set	1	
1.4	Fire Suppression System	Pack and system level aerosol fire suppression, system level combustible gas detectors + explosion-proof ventilation system, water fire suppression system	set	1	
1.5	Container	Dimensions: 6058 * 2896 * 2438mm (W*H*D) Anti-corrosion grade: C5; IP level: IP55	pcs	1	
2	MV Skid Container	2.5MW	set	1	NR
2.1	PCS	PCS-9567-2500 IP level: IP65	set	1	
2.2	Step-up Transformer	2625kVA/ 33-0.69kV, oil immersed	set	1	
2.3	Ring Main Unit	SF ₆	set	1	
2.4	Auxiliary Transformer	125kVA/0.69KV-0.4KV, dry-type	set	1	For battery container and MV skid
2.5	Container	Dimensions: 6058*2896*2438mm(W*H*D),20ft Including internal cable accessories. Anti-corrosion grade: C4-M; IP level: IP54	pcs	1	
3	MV Skid Container	10MW	set	1	NR
3.1	PCS	PCS-9567-2500 IP level: IP65	set	4	
3.2	Step-up Transformer	10300kVA/ 33-0.69kV, oil immersed	set	1	

3.3	Ring Main Unit	SF ₆	set	1	
3.4	Auxiliary Transformer	300kVA/0.69KV-0.4KV, dry-type	set	1	For battery container and MV skid
3.5	Container	Dimensions: 12192*2896*2438mm(W*H*D),40ft Including internal cable accessories. Anti-corrosion grade: C4-M; IP level: IP54	pcs	1	
4	EMS	Energy management system only for BESS	set	1	NR

Note:

1. Equipment and cables outside the scope of supply are not included; inter-container connection cables are specifically excluded from the supply scope.
2. This proposal is configured based on a nominal capacity of 12.5MW/50.15MWh. The design can be adjusted accordingly if there are any changes to the project capacity or requirements at a later stage.
3. Product applicable altitude: <2000m; Ambient temperature range: -5°C~45°C; The rated temperature of the PCS and transformer is 45°C. They will be derated above 45°C, which meets the project requirements. Battery container Anti-corrosion grade:C5, MV station Anti-corrosion grade: C4-M. If the project site is located in a coastal area, the corrosion protection level of the BESS equipment shall be adjusted accordingly.
4. This solution is designed to satisfy the grid-following mode only. If you have grid forming or black start requirements, please contact us.
5. In current solution, EMS controls BESS only.
6. The final configuration and supplier brand selection of the system shall be based on the final design liaison evaluation. This configuration plan is for preliminary reference only.